Project Planning & Management

Stage 1 • ML Diploma Graduation Project • Deliverable Template v1.0

Team Skills Dynamix | Team No. 178 | Group ALX5_AIS2_S1 | Version v1 | 16 October 2026

Project metadata	Details
Team name	Skills Dynamix
Project title	AI Construction Safety Monitor
Submission date	16 October 2026
Version	v1

1. Project Summary

Construction sites contain changing hazards that are difficult to monitor continuously through manual observation alone. Skills Dynamix proposes a Computer Vision-based prototype that processes supported camera/video input, detects selected workers and personal protective equipment (PPE), applies explicit safety rules, assigns configurable incident severity, and records events for review in a web dashboard. The intended users are site safety supervisors and project staff who need a traceable view of detected incidents. The system is a decision-support prototype, not a certified safety system, and its coverage depends on camera quality, data, model performance, and validated rules.

2. Team and Roles

Member name	Email	Primary role	Main responsibilities
Omar Mohamed Elsayed Khalil	Omarkhalilibrahim451@gmail.com	Project Team Leader	Coordination, task tracking, GitHub management, instructor contact, and overall integration.
Naira Wahed Ahmed Ali	nairawahed2@gmail.com	Computer Vision / AI	Dataset preparation, object detection, selected safety-event detection, and model evaluation.
Noureen Hamdy Mahmoued	noureenhamdy3@gmail.com	Frontend & Dashboard	React dashboard, alert and incident views, UI design, and frontend-backend integration.
Seifelden Mohamed Awad Farfour	Seiffarfour@gmail.com	Backend & Database	FastAPI services, incident/alert APIs, MySQL database, and backend integration.
Begad Ahmed Gad Ahmed	begoahmed06@gmail.com	Testing & Integration	System testing, safety-rule validation, performance testing, documentation, and deployment support.

3. Objectives and Success Criteria

Targets below are proposed evaluation targets for team/instructor confirmation, not achieved results. They should be revisited after the dataset and baseline are selected.

ID	Objective	How it is measured	Target
O-01	Prepare a reproducible, labelled dataset for the selected detection classes.	Document source, license, class map, train/validation/test split, and annotation checks.	100% of selected classes have documented definitions; no train/test leakage identified in the final split.
O-02	Train and evaluate a baseline detector for supported person/PPE classes.	Per-class precision, recall, mAP@0.5, mAP@0.5:0.95 on a held-out test set.	Proposed: mAP@0.5 ≥ 0.75 for agreed baseline classes; confirm feasibility after baseline evaluation.
O-03	Integrate detection output with safety rules, incident API/database, and dashboard.	End-to-end test cases for a supported violation, event creation, persistence, and display.	All agreed critical-path acceptance tests pass; no known blocking integration defects at feature freeze.
O-04	Measure usability and runtime performance under a documented test setup.	Measure processing latency/FPS, alert-to-dashboard delay, and test outcomes on specified hardware/video.	Report measured values and hardware; proposed target ≥ 5 processed FPS on the agreed test setup, subject to confirmation.

4. Scope

In scope — planned core	Out of scope / conditional / future
Supported local video or camera stream input; frame processing using Python/OpenCV; baseline YOLO-family object detection for selected classes.	Certified life-safety use, guaranteed detection, automatic enforcement, or replacement of trained safety staff.
Selected PPE detection and configurable rules for supported zones; event severity based on documented rule table.	Reliable person-to-helmet/vest association unless explicitly designed and validated; uncalibrated physical-distance estimates from pixels alone.
FastAPI endpoints, MySQL incident records, React dashboard for alerts/history, and documented setup.	Guaranteed fall detection, proximity-to-equipment risk, or crowd alerts unless suitable labels, methods, and acceptance tests are available.
Offline video demonstration and measurable model/system testing; Git-based collaboration after setup.	Production deployment, multi-site scaling, cloud hosting, advanced tracking/pose estimation, and formal compliance certification.

5. Work Breakdown Structure

ID	Task / deliverable	Owner	Start	End	Depends on
T-01	Confirm scope, use cases, task ownership, and submission plan.	Omar	Week 1	Week 1	—
T-02	Survey datasets; define classes, licensing checks, and annotation plan.	Naira	Week 1	Week 1	T-01
T-03	Create dashboard wireframe and navigation map.	Noureen	Week 1	Week 1	T-01
T-04	Draft MySQL schema for users, incidents, alerts, and risk records.	Seifelden	Week 1	Week 1	T-01

ID	Task / deliverable	Owner	Start	End	Depends on
T-05	Define safety scenarios, acceptance criteria, and initial test cases.	Begad	Week 1	Week 1	T-01, T-02
T-06	Prepare/split data and train baseline detector; document environment.	Naira	Week 2	Week 2	T-02
T-07	Build React dashboard shell and incident/alert components.	Noureen	Week 2	Week 2	T-03
T-08	Implement FastAPI skeleton and database connection/API contracts.	Seifelden	Week 2	Week 2	T-04
T-09	Implement supported safety rules, risk mapping, and event payload contract.	Naira	Week 3	Week 3	T-05, T-06
T-10	Connect alert/event persistence and dashboard views end to end.	Seifelden + Noureen	Week 3	Week 3	T-07, T-08, T-09
T-11	Run integration/performance tests; triage false positives and missed events.	Begad	Week 4	Week 4	T-09, T-10
T-12	Feature freeze, final evaluation, README/docs, demo, report and slides.	Omar (coordination); all owners	Week 4	Week 4	T-11

Schedule note: the proposal describes a four-week plan. The official DEPI stage dates were not present in the supplied proposal/template files, so calendar deadlines remain [To be confirmed]. The week labels above are relative planning windows, not official submission dates.

6. Milestones

Milestone	Date	Deliverable	Owner
Stage 1: Project Planning & Management	[Official DEPI date to confirm]	This document	Omar
Stage 2: Literature Review	[Official DEPI date to confirm]	Literature review document	Naira (research coordination); team review
Stage 3: Requirements Gathering	[Official DEPI date to confirm]	Requirements document	Omar + Begad
Stage 4: System Analysis & Design	[Official DEPI date to confirm]	Architecture/design document	Omar + Seifelden
Internal: data pipeline complete	End of Week 1 / confirm calendar date	Selected data, class map, split and data notes	Naira
Internal: baseline model trained	End of Week 2 / confirm calendar date	Baseline weights and evaluation report	Naira

Milestone	Date	Deliverable	Owner
Internal: feature freeze	Start of Week 4 / confirm calendar date	Feature list frozen; only fixes, tests and documentation thereafter	Omar; all members
Stage 5: Implementation	[Official DEPI date to confirm]	Repository + implementation document	Omar + all owners
Stage 6: Final presentation, testing & reports	[Official DEPI date to confirm]	Report, slides, demo	All; Omar submits

7. Tools and Collaboration

Purpose	Tool / status	Link
Source code repository	Git + GitHub — Pending setup	[To be added after repository is created and access verified]
Task board	Pending setup	[To be added after board is created]
Shared documents	Shared team folder/platform — to be confirmed	[To be added]
Team communication	Team-approved channel — to be confirmed	[To be added]
Experiment tracking (optional)	CSV/Markdown experiment log initially; tool TBD	[Optional / to be added]

The team leader should verify that repository invitations and the team number are correct, confirm each member's access, and submit required deliverables through the LMS. No invitation or access is assumed to have been received or accepted.

8. Communication Plan

Meeting or report	Proposed frequency	Participants and purpose
Team sync	Twice weekly during the four-week plan	All members; review blockers, dependencies, task status and next steps.
Progress update to instructor	At each official stage submission and when a major blocker affects a deadline	Omar as single point of contact; summarize completed work, risks and pending decisions.
Code/design review	At least once per week and before integration	Relevant owner plus one reviewer; check reproducibility, API contracts, tests and documentation.

These are proposed working practices; the team has not confirmed that these meetings or channels are already scheduled.

9. Risk Register

Risk	Likelihood	Impact	Owner	Mitigation
Insufficient or unsuitable labelled construction-safety data	H	H	Naira	Audit class coverage and license early; start with a narrow class set; document

Risk	Likelihood	Impact	Owner	Mitigation
				exclusions and use a held-out test set.
False positives or missed hazards, especially small/occluded PPE	H	H	Naira + Begad	Review per-class precision/recall; inspect errors across lighting/distance/occlusion; make alerts reviewable and document limitations.
Limited compute resources or slow inference	M	M	Naira	Start with a lightweight YOLO variant; measure latency on agreed hardware; reduce frame sampling/resolution only with documented trade-offs.
Backend/frontend/model integration delays	M	H	Omar + Seifelden + Noureen	Agree payload/API contracts in Week 1; integrate a stub early; run end-to-end tests by Week 3.
GitHub/task-board setup or invitation/access delays	M	M	Omar	Create and verify repository/board promptly; check member identity/team number; retain a local backup until access works.
Insufficient time for testing, documentation or final demo	M	H	Omar + Begad	Set feature freeze at start of Week 4; prioritize critical path; reserve final week for fixes, evidence, documentation and rehearsed demo.
Team member unavailable or workload imbalance	M	H	Omar	Keep tasks and handover notes visible; assign reviewers; reallocate tasks transparently without changing official ownership records silently.

10. Definition of Done

- Each task produces the stated deliverable, is reviewed by at least one teammate, and is recorded against its task ID.
- Code runs from documented setup instructions in a clean environment as far as dependencies and available hardware allow; configuration secrets are not committed.
- Model evaluation reports the dataset source, class definitions, split, metrics, hardware, known failure cases and limitations. No unmeasured performance claim is presented as a result.
- Supported safety rules have documented inputs, thresholds, severity mapping and corresponding tests. Unsupported/optional event types are clearly labelled.
- API/database/dashboard integration is exercised by end-to-end tests; incident records include enough context for review and can be viewed in incident history.
- README/setup guide, requirements, testing report, project report, slides and demo assets are reviewed and prepared for LMS submission.
- Feature freeze occurs before the final testing/documentation period; changes after freeze are limited to approved fixes, testing and documentation.

- Repository and task-board links are added only after setup and access are verified. Omar confirms LMS submission; no pending item is marked complete without evidence.

Submission Checklist

- PASS — Every member has a role and owns at least one WBS task.
- PASS WITH QUALIFICATION — Every objective has a measurable proposed target; numerical targets require team/instructor confirmation.
- UNRESOLVED — Milestones include a feature-freeze point, but official calendar dates are not present in supplied materials.
- UNRESOLVED — Repository and task board do not yet exist / have not been confirmed; instructor access is pending.
- PASS — Risk register contains more than four risks, each with mitigation.